SOFTWARE CHANGE IMPLEMENTATION REPORT

CR001-CR004: Library Management System Enhancements

Field	Details
Change Request IDs	CR001, CR002, CR003, CR004
Project	Library Management System
Student Name	V. Susmitha
Student ID	JFICT24_30
Implementation Status	Implemented; targeted verification completed
Report Date	29 October 2026

1. Change Request Summary

1.1 Business Requirement

This report documents four enhancements to the Library Management System: student sign-in and personal borrowing/hold history (CR001); overdue fine calculation and reporting (CR002); search and status filtering of borrowing records (CR003); and overdue highlighting in the borrowing list (CR004).

The changes extend the existing librarian workflows while restricting student access to the signed-in student's own information. The fine rate implemented for CR002 is LKR 10 per overdue calendar day.

1.2 Existing Behaviour

- The system provided librarian sign-in and librarian management functions.
- Students could place holds through the catalogue, but had no authenticated account page for personal borrowing and hold history.
- Borrowing records had due dates and return status, but no persisted overdue fine.
- The borrowing list had no combined search/status filters and no distinct overdue row indicator.

1.3 Requested Change

- Provide student login and a private page showing the logged-in student's borrowings and holds; protect student and librarian routes.
- Calculate fine as overdue calendar days multiplied by the configured daily rate, persist it on return, and display it in borrowing reports.
- Search borrowing records by student name or book; filter by Issued or Returned; allow both controls together and show an empty state.
- Highlight only issued, unreturned records past their due date in red and show an Overdue badge.

2. Implementation

2.1 Implementation Summary

CR001 adds a student login flow using the student's index number and a password hash. Librarians set or reset the password in the student add/edit forms. Authenticated students see only their own borrowing and hold records. Catalogue and hold operations now require a signed-in user; student hold ownership is derived from the session and checked on confirmation/cancellation.

CR002 adds a DECIMAL fine field and a single configuration value of LKR 10 per overdue day. The return confirmation calculates a non-negative fine using the database's current date, displays the amount before confirmation, and saves the return date, fine, and Returned status together. Monthly, student, and book reports include the saved fine.

CR003 adds prepared-query filters for student name, book title/book ID, and the Issued/Returned status. CR004 uses the due date and Issued status to add a red row style and Overdue badge; returned records are not classified as currently overdue.

2.2 Affected Files

File	Change	Purpose / Change Request
public/student-login.php	Added	CR001 student credential login and account guidance.
public/student-dashboard.php	Added	CR001 personal borrowing, fine, and hold lists.
includes/student_auth.php	Added	CR001 student-session guard for student-only routes.
includes/auth.php	Modified	CR001 prevents student sessions entering librarian-only pages.
public/login.php	Modified	CR001 preserves librarian login and links to student login.
public/students/add.php	Modified	CR001 creates student password hashes.
public/students/edit.php	Modified	CR001 allows librarian password reset.
public/index.php	Modified	CR001 authenticated catalogue, student navigation and hold controls.
includes/navbar.php	Modified	CR001 adds catalogue navigation for librarian sessions.
public/hold/create.php	Modified	CR001 associates a hold with the authenticated student.
public/hold/confirmation.php	Modified	CR001 restricts confirmation to its owner.
public/hold/holdcancel.php	Modified	CR001 restricts cancellation to the owning student.
public/logout.php	Modified	CR001 returns logged-out users to Admin Login, not a guest catalogue.
config/settings.php	Added	CR002 fine rate and currency configuration.
schema.sql	Modified	CR001 password_hash and CR002 fine_amount for new databases.
migrations/20261029_cr001_cr002.sql	Added	CR001/CR002 upgrade for an existing database.
public/borrowings/return.php	Modified	CR002 fine preview and persistence on return.
public/reports/monthly.php	Modified	CR002 fine column in monthly report.
public/reports/student.php	Modified	CR002 fine column in student report.
public/reports/book.php	Modified	CR002 fine column in book report.
public/borrowings/index.php	Modified	CR003 search/status filtering and CR004 overdue row/badge.
assets/css/style.css	Modified	CR004 accessible overdue row and badge styling.
README.md	Modified	Documents authenticated catalogue and database migration/fine rate.

2.3 Technical Changes

A. CR001 - Student Authentication and Personal Records

- Student passwords are stored with PHP password hashing; the index number is the login identifier.
- Session regeneration follows successful authentication. Role-aware guards separate student pages from librarian pages.
- Personal borrowing and hold queries are restricted by the authenticated student ID. Hold creation, confirmation, and cancellation check ownership.
- The catalogue redirects unauthenticated visitors to Admin Login; student navigation shows My Records, Book Catalogue, and Logout.

B. CR002 - Overdue Fine Calculation

- Fine storage uses DECIMAL(10,2) with a zero default. The existing database migration adds the student password hash and fine column.
- Fine rate is configured as LKR 10 per overdue calendar day in config/settings.php.
- Fine is max(0, overdue days) multiplied by the daily rate. The saved return date, fine, and status are updated together.
- The fine is shown in monthly, student, book, and student-account borrowing views.

C. CR003 - Borrowing Search and Filter

- Search supports student name, book title, and book ID using prepared parameters.
- Status is restricted to All, Issued, or Returned. Search and status parameters work together.
- A clear message is rendered when no records match.

D. CR004 - Overdue Visual Indicator

- An item is overdue only when its status is Issued and its due date is earlier than the current date.
- Overdue rows receive a red-tinted background and an Overdue badge; returned records are excluded.

2.4 Implementation Logic

1. Authenticate a student and retain the student ID and role in the session.
2. Require the correct role before serving student or librarian functions; use session identity to scope personal data and hold actions.
3. On return, obtain the database date, calculate overdue days/fine, then persist the return data in a transaction.
4. Read the search/status inputs, validate status, execute the prepared borrowing query, and render results or the empty state.
5. For each borrowing row, compare status and due date and render the overdue classes only when both conditions identify an active overdue item.

2.5 Error and Data Handling

- Invalid student credentials are rejected. Students without a password need a librarian to set/reset one.
- Unauthenticated catalogue and hold access redirects to the appropriate login page. Hold access is limited to the owning student.
- Fine calculation is clamped at zero for early/on-time returns.
- Search status values outside the supported set are rejected and reset to All.
- The migration was applied to the configured library_management3 database. Temporary test records used for database acceptance checks were rolled back.

3. Testing & Verification

3.1 Testing Approach

Targeted PHP syntax checks, local MySQL integration checks, fine/date boundary checks, and local HTTP redirect checks were performed. Database acceptance tests used temporary records inside a transaction, then rolled back all test records.

- Environment: Windows development workstation; PHP 8.0.28 CLI; local MySQL database library_management3.
- Test scope: PHP syntax, schema migration, authenticated-route redirects, fine persistence/report queries, search/status query behavior, and overdue date conditions.
- Not performed: full manual visual/browser sign-in, all student/admin regression flows, and responsive layout review.

3.2 Test Cases

Test ID	Test Case	Expected Result	Status
TC01	Run PHP syntax checks on changed PHP pages and guards.	No syntax errors.	Pass
TC02	Inspect migrated students and borrowings tables.	password_hash and fine_amount columns exist.	Pass
TC03	Submit valid and invalid student credentials through the browser.	Valid login opens own account; invalid login is rejected.	Not Tested
TC04	Request catalogue without authentication.	HTTP 302 redirect to Admin Login.	Pass
TC05	Request hold creation without student authentication.	HTTP 302 redirect to Student Login.	Pass
TC06	Check fines for early, due-today, one-day-late, and three-days-late returns.	LKR 0, 0, 10, and 30 respectively.	Pass
TC07	Persist a one-day late return and query monthly, student, and book report datasets.	LKR 10 is stored and available in each report query.	Pass
TC08	Search a temporary student name and filter Issued.	Both matching issued records are returned.	Pass
TC09	Search a temporary book title and filter Returned with no returned match.	Zero matching rows; empty state is provided.	Pass
TC10	Compare issued records due yesterday and due today.	Only the past-due issued record is overdue.	Pass
TC11	Check past-due Returned and future-due Issued records.	Neither is identified as overdue.	Pass
TC12	Sign in as one student and attempt to access another student's records/hold.	Other student's data is inaccessible.	Not Tested
TC13	Review the catalogue, login, and borrowing pages in a browser at desktop/mobile sizes.	Navigation and layout remain usable.	Not Tested
TC14	Run librarian login and existing books, students, issue/return, and reports regression.	Existing librarian workflows continue to work.	Not Tested

3.3 Test Summary

Test Result	Count
Total Test Cases	14
Passed	10

Test Result	Count
Failed	0
Not Tested	4
Pass Rate (executed tests)	100% (10/10)
Execution Coverage	71.4% (10/14)

Overall Test Result: Pending Full Manual Verification. Ten targeted tests passed. Four browser/authentication/regression cases remain untested; the report does not mark those acceptance checks as passed.

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification	Status
AC001	Students can log in with credentials set by the librarian.	Implementation and syntax checked; end-to-end browser login not tested.	Pending
AC002	Students can view only their own borrowing and hold records.	Queries scoped by session student ID; cross-account test pending.	Partial
AC003	Guests cannot open the book catalogue.	Local HTTP request returns 302 to Admin Login.	Pass
AC004	Hold creation, confirmation, and cancellation are student-authenticated and owner-scoped.	Route guards and owner checks implemented; cross-user browser test pending.	Partial
AC005	Fine is zero on/before due date and rate x overdue days after due date.	Early, due-today, 1-day and 3-day boundary cases passed.	Pass
AC006	Fine is saved on return and visible in relevant reports.	Isolated MySQL transaction verified save and three report queries.	Pass
AC007	Search and Issued/Returned filters combine and no-match result is clear.	Prepared query tested with isolated records; visual page test pending.	Partial
AC008	Only unreturned overdue borrowings receive overdue status.	Past-due, due-today, returned, and future-due cases checked.	Pass
AC009	Overdue records display a legible red row and Overdue badge.	Conditional and styles present; visual browser verification pending.	Partial
AC010	Existing librarian login and management workflows remain functional.	Regression test not performed.	Pending
AC011	Existing database is upgraded with required columns.	Migration applied and both columns confirmed.	Pass

5. Evidence

The following verification evidence was collected during implementation. Screenshots are not attached; visual browser evidence should be added after the remaining manual tests.

Evidence 01 - Access Control Redirects

Local HTTP checks verified an unauthenticated catalogue request redirects to Admin Login (HTTP 302), and unauthenticated hold creation redirects to Student Login (HTTP 302).

Evidence 02 - Isolated Database Acceptance Checks

Temporary records were used to verify one-day overdue fine persistence at LKR 10 and the monthly, student, and book report queries. Search/status filters and overdue date classification were also checked. The test transaction was rolled back after verification.

Evidence 03 - Syntax and Migration

PHP lint checks passed for the changed PHP pages. The migration added students.password_hash and borrowings.fine_amount to the configured database. The diff whitespace check passed.

6. Final Result

Item	Result
Requirements	Implemented
Implementation	Implemented
Targeted automated/database tests	Passed
Manual browser testing	Pending
Acceptance criteria	Partially verified
Data integrity	Temporary test records rolled back
Final Status	Implemented - Pending full manual verification

6.1 Final Summary

CR001-CR004 have been implemented in the Library Management System. Students have a role-separated login and private account view; fine amounts are calculated and retained on returns; borrowing search and status filters are available; and overdue issued rows receive a visual indicator. Fine reporting is available in monthly, student, and book reports.

Targeted syntax, schema, redirect, database query, fine calculation, and overdue-condition checks passed. Full student credential and cross-student authorization flows, visual presentation, and librarian regression workflows still require manual verification. Therefore, implementation is complete, but final acceptance testing remains pending.

6.2 Issues / Limitations

- Existing students must have a password set/reset by a librarian before their first sign-in. Student self-registration is not included.
- The configured fine policy is LKR 10 per overdue calendar day.
- The migration has already been applied to the configured local library_management3 database. Run it once in any other existing database before using the changes.
- Complete the four Not Tested cases and attach browser screenshots before marking final acceptance as verified.

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